

MUHAMMAD ZAFRAN SYAUQI BIN BADDERISAH

Electronics Engineer | Embedded Systems | C/C++ | MCU/FPGA | Hardware Design | PCB | Testing & Debugging

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PROFESSIONAL SUMMARY

Fresh graduate Electronics Engineer with hands-on experience in robotics-related systems, embedded systems, motor control, hardware integration, testing and troubleshooting. Developed an ESP32-based rehabilitation system integrating BLDC motor actuation, AS5600 encoder feedback, FOC, PID control and mechanical assemblies into a functional closed-loop prototype. Experienced in testing, calibration, debugging and analysing system performance, supported by manufacturing QA/QC and electronics troubleshooting experience at Panasonic. Strong foundation in C/C++, sensor integration, control systems, electrical hardware, and system integration, with a strong interest in developing practical automation and robotic systems.

EDUCATION

Universiti Tun Hussein Onn Malaysia (UTHM)

Bachelor of Electronic Engineering with Honours | March 2023 – July 2026

Universiti Tun Hussein Onn Malaysia (UTHM)

Diploma in Electrical Engineering | August 2020 – January 2023

ACADEMIC PROJECTS

Development of AReS: Accessible Rehabilitation System for Forearm Recovery

Final Year Project | Embedded Systems & Real-Time Control

Winner: IEEE Control Systems Society Award, InnoCube 3D LAB Special Award, Gold Award & Most Outstanding FYP

- Engineered an ESP32-based automated rehabilitation system integrating BLDC motor actuation, AS5600 magnetic encoder feedback, and closed-loop position control.
- Implemented FOC and PID control algorithms in C/C++ to regulate motor position and dynamic resistance based on real-time sensor feedback.
- Integrated electrical, mechanical, and embedded subsystems, using Autodesk Fusion 360 to design mechanical assemblies for the rehabilitation prototype.
- Tested and debugged motor-control performance by calibrating encoder feedback and control parameters to address vibration, position error, and thermal overshoot.
- Developed real-time data logging to monitor and analyse range-of-motion performance during system operation.

Smart Street Light System Using Zynq-7000 SoC FPGA

- Architected an adaptive street-lighting control system on a Zynq-7000 SoC FPGA to process environmental inputs and automate lighting control.
- Designed digital logic and embedded processing workflows for sensor-driven lighting decisions and energy-efficient operation.

Energy Saving Solar Power for Microalgae Farm

Integrated Design Project

- Designed a solar-powered energy architecture for a microalgae farming facility, covering system requirements, energy generation, storage, and distribution.
- Evaluated PV panels, battery storage, and power-distribution requirements to develop an energy-efficient system architecture.

ONN THE GO- UTHM Student Services Hub Mobile App

Software Project

- Developed an Android application prototype using Android Studio to centralize access to university student services.
- Designed front-end navigation and user workflows using UI/UX principles and SDLC practices to improve service accessibility.

PROFESSIONAL EXPERIENCE

MYBKK ENTERPRISE SDN BHD — On-Site at Panasonic Manufacturing Malaysia Berhad

Manufacturing Engineer / QA-QC Intern | July 2024 – October 2024

- Executed QA/QC testing and hardware failure analysis on PCB assemblies and heater-tank systems within a continuous manufacturing environment.
- Diagnosed PCB defects using component-level electrical measurements and digital multimeters to isolate hardware failures and identify root causes.
- Investigated production hardware deviations and collaborated with production operators and cross-functional teams to identify failure mechanisms and recommend corrective actions.
- Supported manufacturing quality improvement by translating troubleshooting findings into technical feedback for production teams.

AMY'S COMPUTER SDN BHD

Computer & Electronics Repair Technician Intern | August 2022 – January 2023

- Diagnosed and repaired electronic hardware faults across laptop and desktop systems through systematic inspection, electrical testing, fault isolation, and functional verification.
- Troubleshoot power-delivery circuits, BIOS failures, wiring faults, and degraded board components using electrical measurements and structured hardware-debugging procedures.
- Performed component-level fault isolation and hardware diagnosis to restore system functionality and reduce equipment downtime.
- Replaced and configured computer hardware components including RAM and SSDs while validating system operation after repair.

CORE SKILLS

Embedded Systems & Programming: C/C++, ESP32, Arduino UNO, Zynq-7000 SoC FPGA, Python, MATLAB

Electronics & Hardware: PCB Testing, PCB Layout, Component-Level Troubleshooting, Electrical Measurements, Hardware Failure Analysis, Sensor Integration, Encoder Feedback

Control Systems: BLDC Motor Control, Field-Oriented Control (FOC), PID Tuning, Closed-Loop Control, PWM, I2C, SPI, UART, SimpleFOC

Product Development: Hardware Prototyping, System Integration, Testing & Validation, Root-Cause Analysis, Mechanical Integration

CAD & Engineering Tools: Autodesk Fusion 360, KiCad, Proteus, Xilinx Vivado, VS Code, Arduino IDE

Office Tools: Microsoft Excel, PowerPoint, Word

AWARDS & ACHIEVEMENTS

Most Outstanding Final Year Project Award (Electronic Engineering) – UTHM | 2026

Gold Award for Final Year Project – UTHM | 2026

InnoCube 3D LAB Special Award for Final Year Project – InnoCube | 2026

IEEE Control System Society Award for Final Year Project – IEEE Control System Society | 2026

Best Volunteer Achievement Award - Faculty of Electrical and Electronic Engineering (FKEE), UTHM | 2026

Sapta Pilar Bakti Award (Excellence in Volunteering) - Kelab Sukarelawan UTHM | 2025

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Kelab Sukarelawan UTHM

Deputy Chairman | 2025 – 2026

- Directed administrative operations, promotional campaigns and club-branding initiatives supporting university-wide volunteer programs.
- Coordinated committee workflows and cross-functional teams to execute volunteer activities against operational requirements and deadlines.

Pejuang Jalanan 2.0: Misi Pantai Timur

Assistant Director | National | July – December 2025

- Directed logistics, resource allocation, and field deployment for a national-level volunteer mission across Malaysia's East Coast region.

Program Bakti Madani UTHM 2025

Head Committee | National | March – May 2025

- Led committee operations and task delegation for a national-level program, aligning workstreams with established timelines and program objectives.